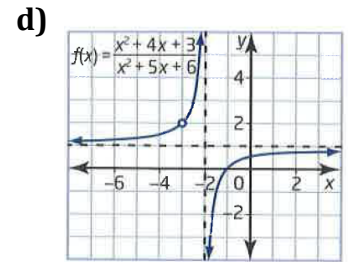
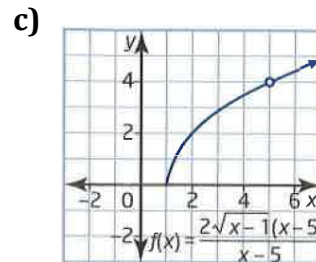
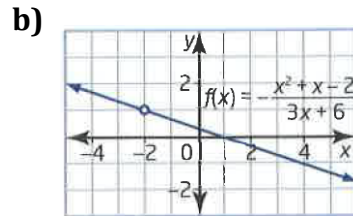
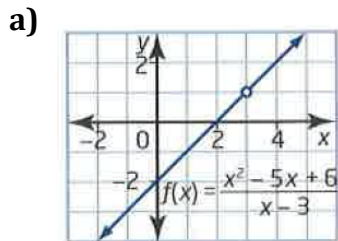


2.1/2.2 Multiplying and Dividing Rational Expressions – Worksheet

MCR3U

1) State the restrictions for each function.



2) Simplify each expression and state all restrictions on x .

a) $\frac{x-8}{x^2-13x+40}$

b) $\frac{x^2-3x-18}{x^2+x-42}$

c) $\frac{x+8}{x^2+6x-16}$

3) Simplify and state the restrictions on the variables.

a) $\frac{14y}{11x} \times \frac{121y}{7x}$

b) $\frac{15b^3}{4b} \times \frac{20b}{30b^2}$

c) $\frac{5x}{9y} \div \frac{5x}{18y^2}$

d) $\frac{26ab}{4a} \div \frac{39a^4b^3}{12b^4}$

4) Simplify and state the restrictions on the variable.

a) $\frac{25}{x+10} \times \frac{x+10}{5}$

b) $\frac{x+5}{x-3} \times \frac{x-3}{x+7}$

$$\text{c)} \frac{x+1}{x} \div \frac{x+1}{2x}$$

$$\text{d)} \frac{x+12}{x+10} \div \frac{x+12}{x-5}$$

5) Simplify and state the restrictions on the variable.

$$\text{a)} \frac{3x^2}{12x^2+18x} \times \frac{4x+6}{3x+30}$$

$$\text{b)} \frac{4x+24}{x^2+8x} \times \frac{12x^2}{3x+18}$$

$$\text{c)} \frac{x^2+10x+21}{x+3} \times \frac{x+2}{x^2+9x+14}$$

$$\text{d)} \frac{x^2+2x-15}{x^2-9x+18} \times \frac{x-6}{x+5}$$

6) Simplify and state the restrictions on the variable

$$\text{a)} \frac{x^2+15x}{4x+24} \div \frac{3x}{3x+18}$$

$$\text{b)} \frac{6x}{8x-72} \div \frac{9x}{2x-18}$$

$$\text{c)} \frac{x^2+15x+26}{6x^2} \div \frac{x^2-3x-10}{30x^3}$$

$$\text{d)} \frac{x^2+11x+24}{x^2+2x-3} \div \frac{x-8}{x-1}$$

7) Simplify and state the restrictions on the variable

$$\text{a)} \frac{a^2-25}{a+2} \cdot \frac{a^2-4}{a^2-7a+10}$$

$$\text{b)} \frac{y^2-4y-21}{3y^2+6y} \cdot \frac{y^2+8y}{y^2+11y+24}$$

$$\text{c)} \frac{p^2-2p+1}{p+1} \div \frac{p^2-1}{p+1}$$

$$\text{d)} \frac{x^2+6x-27}{x^2+11x+18} \div \frac{x-3}{x^2+x-2}$$